
Introduction, Exponential Smoothing & Forecast Evaluation

BSAD 8310: Business Forecasting — Lecture 1

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Lecture Outline

1 Why Forecasting Matters

2 Exponential Smoothing

3 Forecast Evaluation

4 Course Roadmap

Why Forecasting Matters

Every business decision made today depends on expectations about tomorrow.

Forecasting Is Everywhere in Business

Operations & Supply Chain

- How much inventory should Walmart order for Black Friday?
- How many nurses does a hospital need next Tuesday night?
- When will this machine component fail?

Finance & Risk

- Will credit card defaults rise next quarter?
- What will the Fed Funds rate be in 6 months?

Marketing & Strategy

- How many units will this new product sell in year one?
- What is the customer lifetime value of this cohort?

Macro & Public Policy

- Will GDP grow by 2% or 3% next year?
- What will unemployment be in Q4?

A decision is only as good as the forecast it rests on.

The Basic Setup

Time Series

A **time series** is an ordered sequence of observations $\{y_1, y_2, \dots, y_T\}$ where t indexes time.

Key notation:

- y_t — observed value at time t
- h — **forecast horizon** (periods ahead)
- $\hat{y}_{t+h|t}$ — forecast of y_{t+h} made at time t
- $e_t = y_t - \hat{y}_{t|t-1}$ — **forecast error**

Example

Forecasting monthly retail sales:

$h = 1$: next month's sales

$h = 12$: sales 12 months from now

Harder problem as h grows — uncertainty compounds.

Under squared error loss, the optimal forecast is the **conditional expectation**: $\hat{y}_{t+h|t} = \mathbb{E}[y_{t+h} | \mathcal{F}_t]$, where $\mathcal{F}_t$ is all information available at time t .

Components of a Time Series

Any time series can be decomposed (additive form): $y_t = T_t + S_t + C_t + I_t$

- T_t : **Trend** — long-run direction
- S_t : **Seasonality** — regular calendar-driven pattern
- C_t : **Cycle** — medium-run business cycle swings
- I_t : **Irregular** — random shocks

Which Components Matter?

- **Retail sales:** trend + seasonality
- **Quarterly GDP:** trend + cycle
- **Daily stocks:** mostly irregular
- **Energy demand:** all four

Ignoring seasonality causes large, systematic errors. A non-seasonal model on monthly retail data will consistently under-forecast in December and over-forecast in January.

Always Beat a Benchmark First

Before presenting any forecast model, check that it beats a **naïve benchmark**. A model that cannot beat the baseline has zero value.

Standard benchmarks:

Naïve: $\hat{y}_{t+h|t} = y_t$

Seasonal naïve: $\hat{y}_{t+h|t} = y_{t+h-m}$

Mean: $\hat{y}_{t+h|t} = \bar{y}_t$

Drift: $\hat{y}_{t+h|t} = y_t + h\hat{c}$

Which to Use?

- No trend, no season → Mean
- Trend, no season → Drift
- Season → Seasonal naïve
- Asset prices → Naïve

The M4 Competition: many ML models *failed* to beat simple ETS methods (Makridakis et al. 2020).

Exponential Smoothing

Recent observations should carry more weight than old ones.

The Problem with Equal Weights

Most classical methods treat every historical observation equally — but what if the world **changes**?

Business Context

A retailer's sales pattern shifts after a store renovation. Equal-weighting pre-renovation data leads to months of systematic over- or under-forecasting.

Exponential weighting: assign geometrically declining weights so that *recent* observations matter more.

Observation	Age	Weight
y_T	0	α
y_{T-1}	1	$\alpha(1 - \alpha)$
y_{T-2}	2	$\alpha(1 - \alpha)^2$

Parameter $\alpha \in (0, 1]$ controls forgetting rate. High α = short memory; low α = long memory.

Simple Exponential Smoothing (SES)

SES Update Equation

Let ℓ_t be the **level** at time t :

$$\ell_t = \alpha y_t + (1 - \alpha) \ell_{t-1}, \quad 0 < \alpha \leq 1$$

Forecast: $\hat{y}_{T+h|T} = \ell_T$ for all horizons $h \geq 1$ (flat)

Large α (e.g. 0.8):

- Reacts quickly to shocks
- Noisy, volatile forecasts
- Good for rapidly changing series

Small α (e.g. 0.1):

- Slow to adapt
- Smooth, stable forecasts
- Good for slowly evolving series

Special cases: $\alpha = 1 \Rightarrow$ naïve forecast. $\alpha \rightarrow 0 \Rightarrow$ historical mean. SES **nests** both benchmarks.

Use when: no trend, no seasonality.

Holt's Method: Adding Trend

SES produces a *flat* forecast. If the series has a trend, track both level and slope.

Holt's Linear Exponential Smoothing (Holt 2004)

Level: $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$

Trend: $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$

Forecast: $\hat{y}_{T+h|T} = \ell_T + h b_T$

Linear extrapolation can be unrealistic for long horizons. Use the **damped trend** variant for $h > 6$: $\hat{y}_{T+h|T} = \ell_T + \sum_{j=1}^h \phi^j b_T$, where $0 < \phi < 1$ tapers the trend.

Interpretation

ℓ_t : where the series *is* now

b_t : the current growth rate

ϕ : how fast the trend fades

$\phi = 0.85$: forecast converges to a finite ceiling rather than growing forever.

Holt-Winters: Level, Trend, and Seasonality

Holt-Winters Additive Method (Winters 1960)

$$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$$

Forecast: $\hat{y}_{T+h|T} = \ell_T + h b_T + s_{T+h-m}$ (m = seasonal period)

Additive vs. Multiplicative:

- **Additive:** seasonal swings are constant in size
- **Multiplicative:** swings grow with the level (use for retail sales, tourism)

US Retail Sales

The December holiday spike grows each year as overall sales grow $\Rightarrow$ **multiplicative seasonality**.

Requires $\geq 2m$ obs. to initialize ($m = 12$ for monthly).

The ETS Framework: A Unified View

Hyndman and Athanasopoulos (2021) unify all exponential smoothing models as **state space models**: Error $\times$ Trend $\times$ Seasonal.

Component	Options	Meaning
Error (E)	A, M	Additive / Multiplicative
Trend (T)	N, A, A_d	None / Additive / Damped
Seasonal (S)	N, A, M	None / Additive / Multiplicative

15 valid combinations. Common examples:

ETS(A,N,N) = SES

ETS(A,A,N) = Holt linear

ETS(A, A_d ,N) = Holt damped

ETS(A,A,A) = Holt-Winters additive

ETS(A,A,M) = Holt-Winters multiplicative

Automatic selection: fit all 15 models; pick the one with minimum AIC. In Python:
`statsmodels.tsa.exponential_smoothing.ets.ETSModel`

Forecast Evaluation

Forecasting without evaluation is not science — it is storytelling.

How Do We Measure Forecast Accuracy?

Given out-of-sample errors $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$:

Scale-dependent:

$$\text{RMSE} = \sqrt{\frac{1}{H} \sum_{h=1}^H e_{T+h}^2}$$

$$\text{MAE} = \frac{1}{H} \sum_{h=1}^H |e_{T+h}|$$

Same units as y_t . Use to compare models on the *same* series.

RMSE vs. MAE: RMSE penalizes large errors more heavily. If big misses are especially costly (e.g., stockouts), prefer RMSE.

Scale-free (cross-series comparison):

$$\text{MAPE} = \frac{100\%}{H} \sum_{h=1}^H \left| \frac{e_{T+h}}{y_{T+h}} \right|$$

$$\text{MASE} = \frac{\text{MAE}}{\text{MAE}_{\text{seasonal naïve}}}$$

MASE < 1: beats seasonal naïve.

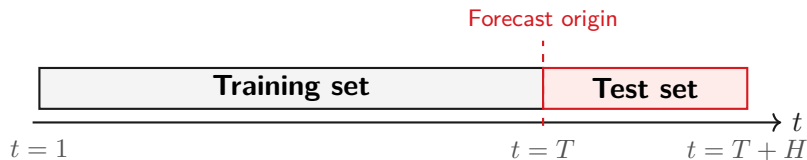
MASE > 1: **worse** than naïve.

Standard in M-Competitions (Makridakis et al. 2020).

The Golden Rule: Always Evaluate Out-of-Sample

Never evaluate a forecast model on the data used to fit it. In-sample fit measures how well the model memorized the past, not how well it can predict the future.

Correct time series train/test split:



Never randomly shuffle time series observations. Always use `TimeSeriesSplit` in scikit-learn, not `KFold`. Shuffling allows future data to “train” the model — a form of data leakage that makes results artificially good.

Course Roadmap

What we will cover, and how it fits together.

Part I: Time Series Models

1. Intro, ETS & Evaluation ← today
2. ARIMA & VAR
3. Generalized Additive Models

Part II: Tree-Based Models

4. Decision Trees
5. Random Forests
6. Boosted Trees & Regularization

Python throughout: statsmodels, scikit-learn, xgboost, PyTorch, prophet, pyGAM, pymc.

Part III: Deep Learning

7. Neural Networks (PyTorch)
8. CNN Architectures
9. RNNs, LSTMs & Transformers

Part IV: Bayesian Models

10. Bayesian Statistics I: Foundations
11. Bayesian Statistics II: Time Series
12. Bayesian Statistics III: Linear Regression

Tools and What You Will Be Able to Do

Python libraries we will use:

- `statsmodels` — ETS, ARIMA, VAR
- `scikit-learn` — ML models, pipelines, CV
- `xgboost` — gradient boosting
- `torch` (PyTorch) — neural networks
- `prophet` / `pygam` — GAMs
- `pymc` — Bayesian models

By the end of this course, you will:

- ✓ Choose the right method for a business problem
- ✓ Implement and tune models in Python
- ✓ Evaluate models rigorously out-of-sample
- ✓ Interpret results for non-technical stakeholders
- ✓ Use AI tools (ChatGPT, Claude) effectively to accelerate analysis
- ✓ Avoid the most common forecasting mistakes

Lecture 1: Key Takeaways

1. Forecasting underpins every forward-looking business decision. The optimal point forecast (under squared loss) is the **conditional expectation**.
2. **Always beat a benchmark first.** Naïve, seasonal naïve, mean, drift.
3. Exponential smoothing assigns **geometrically declining weights** to past observations. α controls how fast old data is forgotten.
4. SES (level only) $\rightarrow$ Holt (+ trend) $\rightarrow$ Holt-Winters (+ season). The **ETS framework** unifies all variants; AIC selects automatically.
5. **Evaluate out-of-sample only.** RMSE penalizes large errors; MASE benchmarks against seasonal naïve. Never shuffle time series data.

Next: ARIMA & VAR — a regression-based approach to time series with rich autocorrelation structure.

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